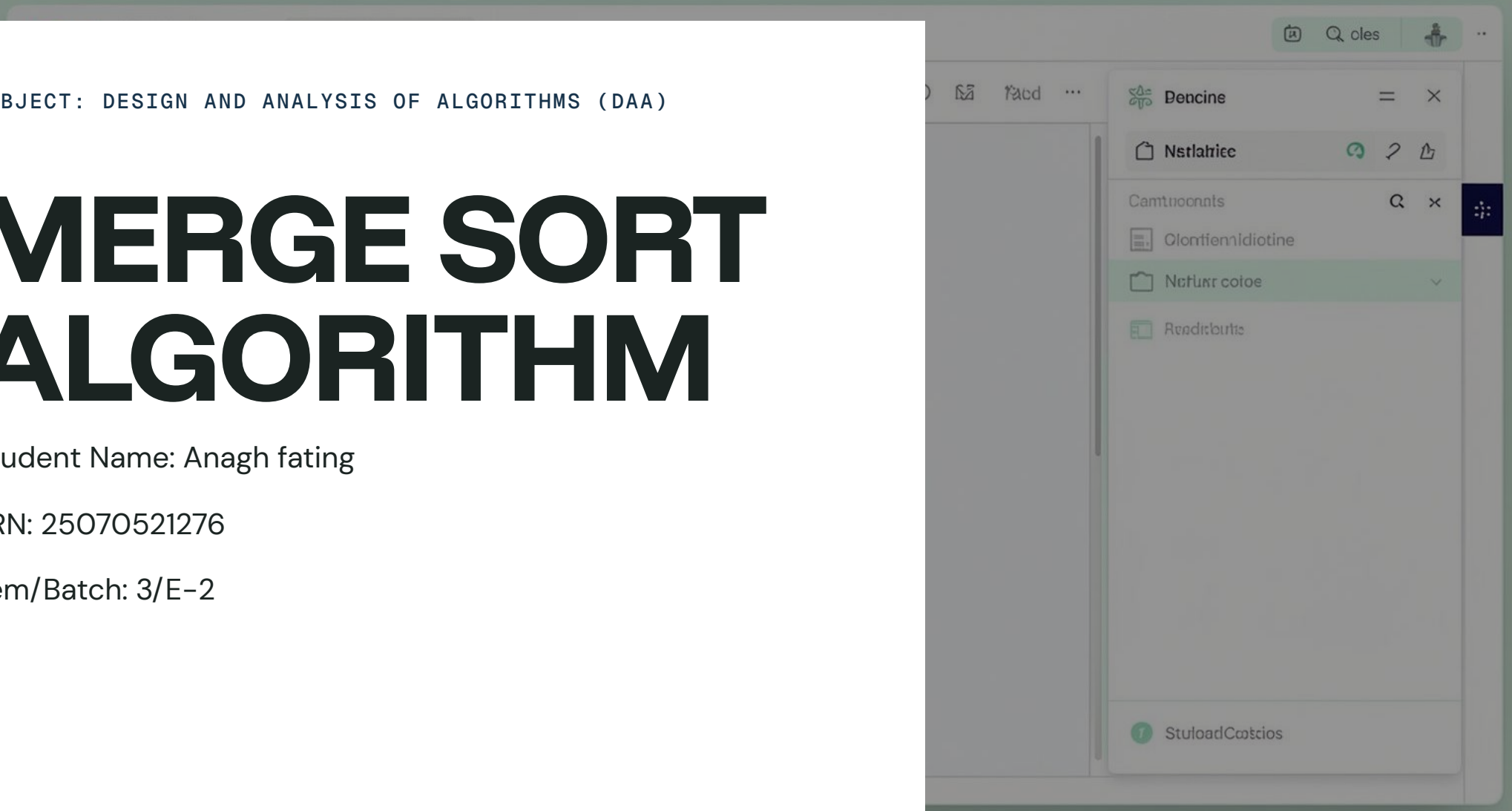
SUBJECT: DESIGN AND ANALYSIS OF ALGORITHMS (DAA)

MERGE SORT ALGORITHM

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sem/Batch: 3/E-2



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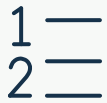
WHY SORT?

ORDER MAKES DATA EASIER TO USE.

A FOUNDATION FOR SEARCHING, GROUPING, AND ANALYSIS.

Sorting organizes data for faster use

Sorting arranges elements in ascending or descending order. Merge Sort is a comparison-based sorting algorithm that follows the Divide and Conquer approach.



Arrange

Put values into a chosen order.



Compare

Use comparisons to decide placement.



Divide

Solve smaller parts before combining them.

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Divide, solve, and combine

Merge Sort repeats these steps until the complete array is sorted.

01

Divide

Split the array into smaller subarrays.

02

Conquer

Sort each subarray recursively.

03

Combine

Merge the sorted subarrays.

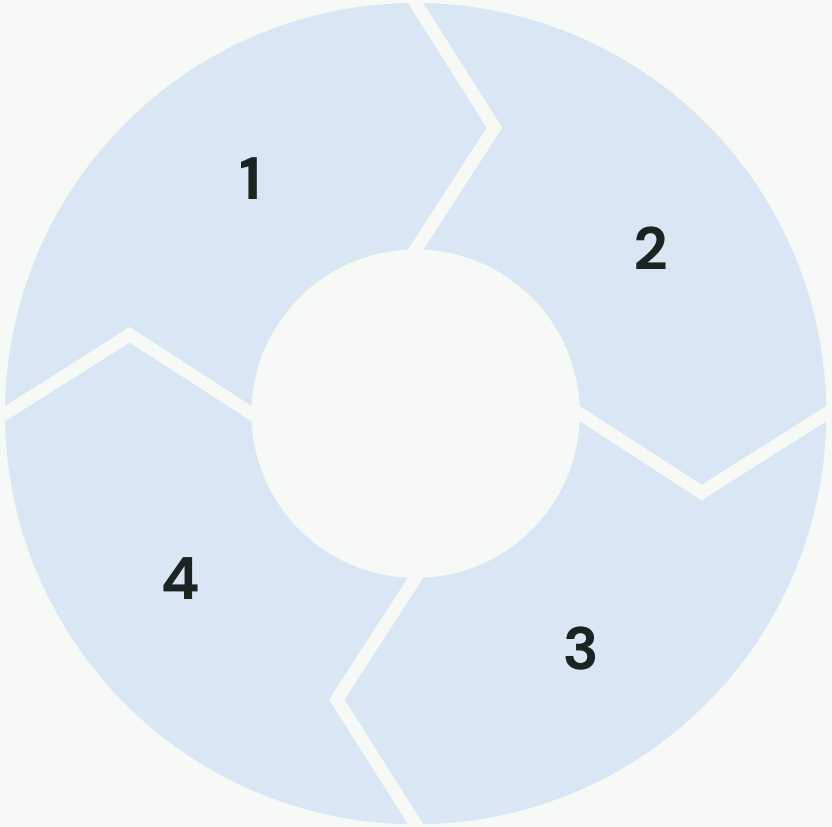
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Merge Sort keeps splitting, then merges in order

Repeatedly divide the array until every part contains one element. A one-element part is already sorted.

Split

Break the array into smaller parts.



Reach one

Each one-element part is sorted.

Merge

Build one ordered array.

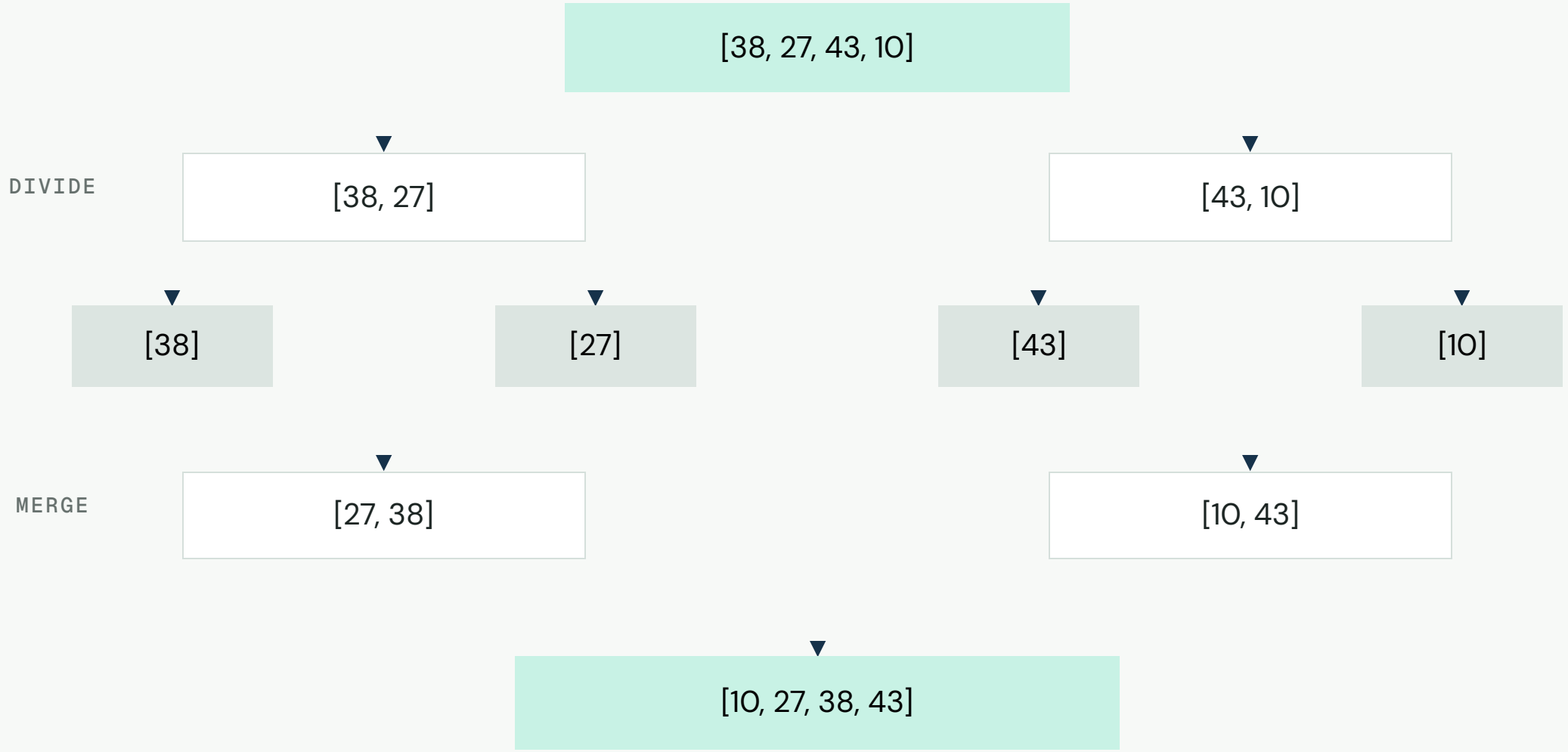
Compare

Check the smaller front element.

CONTINUE MERGING UNTIL ONE SORTED ARRAY REMAINS.

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Example: [38, 27, 43, 10] becomes sorted



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IMPLEMENTATION LOGIC

SPLIT AROUND THE MIDPOINT.

RECURSIVE CALLS HANDLE BOTH HALVES.

Merge Sort pseudocode

```
MergeSort(A, left, right)
  if left < right
    mid = (left + right) / 2
    MergeSort(A, left, mid)
    MergeSort(A, mid + 1, right)
    Merge(A, left, mid, right)
```

Split around the midpoint, sort both halves recursively, then merge the two sorted halves.

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Merge Sort has predictable O(n log n) time



Best case

O(n log n)



Average case

O(n log n)



Worst case

O(n log n)



Auxiliary space

O(n) for the standard array-based implementation



Stable

Equal elements keep their original relative order.

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TAKEAWAY

DIVIDE THE DATA. SORT THE PARTS. COMBINE IN ORDER.

Merge Sort is useful for sorting large datasets and works well for external sorting when data does not fit in memory.

GENERATED USING GAMMMA



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